The Entangled Universe-Anti-Universe Model

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Abstract

We present a cosmological hypothesis in which the universe and an anti-universe emerged simultaneously from a symmetric singularity at the origin of time. This dual creation produced mirror realms with opposite arrows of time, complementary matter–antimatter composition, and entanglement maintained through a pervasive quantum foam. Black holes and white holes act as conjugate portals forming translating wormholes between the two domains, enabling both matter–antimatter transformation and potential time travel without causal paradoxes. The model provides a unified resolution to several standing cosmological puzzles, including the matter–antimatter asymmetry, arrow-of-time directionality, horizon and flatness problems, and the illusion of cosmic expansion. Within the interpretative framework of the Copenhagen and Anthropic principles, this hypothesis proposes that the observed universe represents one half of a globally symmetric, entangled whole.

Keywords: universe-anti-universe, quantum foam, intentional collapse, cosmic fine-tuning, wormholes

1 Introduction

Modern cosmology faces deep conceptual and empirical challenges. Chief among them are the origin of structure from a singular beginning, the apparent dominance of matter over antimatter, the unidirectional flow of time, and the accelerating expansion inferred from redshift data. While inflationary models [1, 2] have addressed some of these concerns, they introduce fine-tuning problems and lack direct observational confirmation.

Quantum cosmology, in turn, offers a probabilistic account of the universe's wavefunction [3, 4], but it remains unclear how a specific classical reality arises from the

quantum ensemble. This paper proposes an alternative: that the Big Bang produced not one universe, but a pair of entangled, complementary universes—a universe and an anti-universe—whose symmetry and entanglement naturally explain the observed asymmetries and dynamic stability of our cosmos.

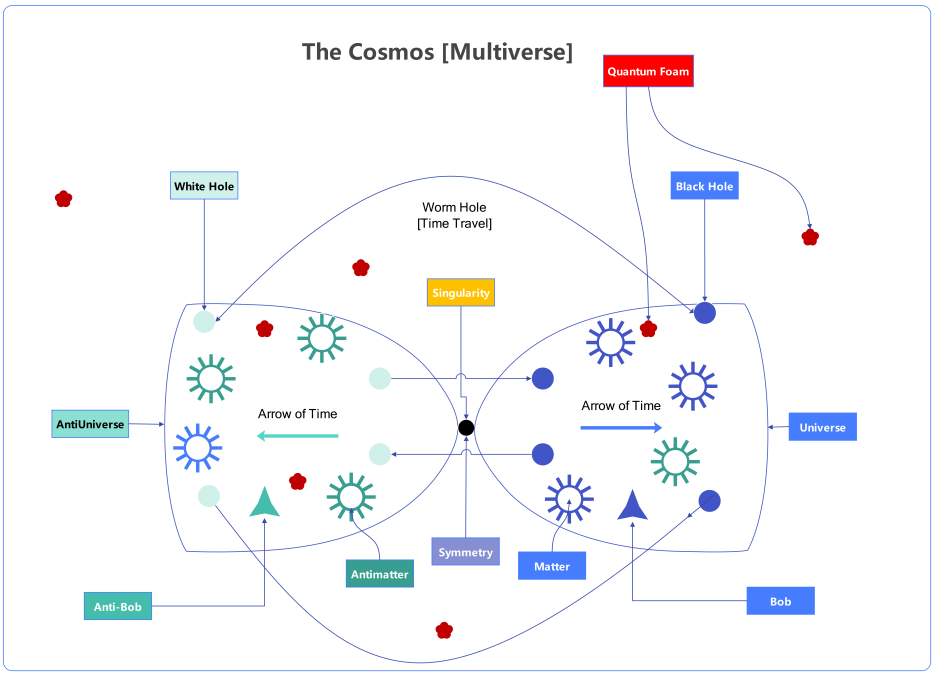

Fig. 1 The Entangled Cosmos

2 Literature Review

The concept of multiple or mirror universes has long been considered in theoretical physics. Sakharov [5] proposed matter–antimatter asymmetry arising from CP violation in the early universe, while Barbour [6] and Penrose [7] examined time-symmetric cosmologies in which entropy increase defines temporal direction. Carroll and Chen [8] further suggested a universe/anti-universe pair evolving in opposite temporal directions from a shared low-entropy boundary.

Quantum gravity approaches, particularly Wheeler's notion of "quantum foam" [9], describe spacetime at the Planck scale as a network of fluctuating wormholes. These ideas support non-local connectivity between separated regions, suggesting a substrate for entanglement even across cosmological scales.

Recent work by Turok and collaborators [10] has advanced the concept of a time-symmetric "anti-universe" preceding ours, emerging from the same Big Bang

singularity with an opposite arrow of time. The present hypothesis builds upon and generalizes these insights, embedding them in a graph-theoretic, non-local framework linking symmetry, entanglement, and topology.

3 Interpretations

Two philosophical interpretations dominate cosmology and quantum theory: the Copenhagen interpretation and the Anthropic Cosmological Principle.

3.1 Copenhagen Perspective

In the Copenhagen view, physical reality manifests only upon measurement, with quantum potentiality collapsing into definite outcomes. Applied cosmologically, the universe's large-scale features may be viewed as the macroscopic realization of underlying quantum superpositions. The singularity, as a total superposed state, requires symmetry breaking to generate observable diversity.

3.2 Anthropic Principle

The Anthropic Cosmological Principle [11] argues that the universe must possess properties compatible with the emergence of observers. Fine-tuning of constants—such as the cosmological constant, gravitational coupling, and entropy at the Big Bang—is often explained anthropically. However, in the entangled pair model, such fine-tuning emerges naturally from global symmetry: parameters balanced across both universes yield stability without invoking probabilistic selection.

4 The Hypothesis: An Entangled Universe–Anti-Universe Pair

We propose that the primordial singularity was a perfectly symmetric, undifferentiated quantum state. Through a spontaneous symmetry break, this state bifurcated into two complementary domains: a matter-dominated universe and an antimatter-dominated anti-universe. These remain entangled through a shared quantum foam, preserving global coherence.

The key features are:

- **Opposite Arrows of Time:** Each universe experiences time moving away from the singularity, but in opposite directions. The singularity is the shared past for both.
- **Matter–Antimatter Complementarity:** Matter dominates one realm, antimatter the other, resolving the apparent asymmetry within a single, symmetric whole.
- **Black–White Hole Symmetry:** Black holes in our universe correspond to white holes in the anti-universe, forming conjugate portals. A black hole absorbs matter and emits antimatter into the anti-universe, maintaining global conservation.
- **Wormholes as Translators:** Black–white hole pairs form traversable wormholes translating between matter and antimatter, enabling potential time-reversal transitions without paradox.

- **Quantum Foam as Substrate:** A pervasive non-local network mediates entanglement and energy exchange, supporting cross-universal coherence.

The universe is not expanding into empty space; rather, the apparent expansion results from the continuous progression of the arrow of time within a fixed dual geometry.

5 Discussion

The entangled pair framework offers a unified explanation for cosmological structure without inflation or exotic energy components. Entanglement through quantum foam provides instantaneous non-local correlation, explaining the uniformity of the cosmic microwave background. The absence of direct observation of the anti-universe is expected, as it exists beyond our temporal domain, though theoretically accessible via wormholes.

The model retains compatibility with general relativity's local curvature description and quantum mechanics' non-local entanglement. Graph-theoretic reasoning shows this system is dynamically stable: the universe and anti-universe serve as balanced nodes connected via foam-mediated edges, ensuring overall conservation and preventing collapse.

6 Questions Resolved

The hypothesis addresses several longstanding puzzles:

- **Origin of Structure:** Emerges from symmetry breaking of the singularity.
- **Matter–Antimatter Asymmetry:** Balanced globally across the entangled pair.
- **Arrow of Time:** Dual opposing arrows explain temporal directionality.
- **Horizon and Flatness Problems:** Non-local entanglement via quantum foam removes the need for inflation.
- **Dark Energy:** Apparent expansion arises from temporal progression, not spatial stretching.
- **Information Paradox:** Black–white hole conjugation preserves information globally.

7 Open Questions

Despite its coherence, several open issues remain:

- **Mechanism of Symmetry Breaking:** The physical trigger for bifurcation requires formalization.
- **Energy Accounting:** Conversion between matter and antimatter across wormholes must respect local conservation.
- **Quantum Foam Dynamics:** Stability under extreme conditions near singularities is not yet quantified.

- **Observational Signatures:** Possible measurable consequences include anomalous neutrino asymmetries, gravitational wave imprints, or subtle correlations in cosmic background radiation.

8 Conclusion

The Entangled Cosmic Creation Hypothesis provides a cosmological framework that unifies quantum non-locality, symmetry, and relativity into a single coherent model. By interpreting the universe and anti-universe as entangled complements originating from a symmetric singularity, it offers pragmatic resolutions to persistent cosmological problems while preserving empirical consistency with known physics. Future research should focus on mathematical formalization of foam-mediated entanglement and potential observational tests of conjugate cosmic structures.

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